

Universal Software-Driven Constraint Tracking Form

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Purpose:

- This document captures constraints originating from Synkly software architecture, including performance targets, reliability requirements, security policies, privacy boundaries, and resource limits.
- These constraints apply to the Synkly web application for student schedule matching and relate to system performance, availability, data protection, and database capacity.
- Traceability to IEEE 29148 (SRS), GDPR/KVKK (privacy), ISO/IEC 25010 (quality), ISO 27001 (security) and other domain-specific standards is enabled where required.

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Constraint ID	Category	Constraint Description	Design Ratio	Owner	Impacted Requirements	Domain Application(s)	Impacted Components	Associated Standard(s)	Verification Method	Risk Impact	Status
SWC-01	Performance	Free time matching $\leq 3s$ for ≤ 100 friends.	Improve response time.	Backend Dev	REQ-PERF-01, REQ-MATCH-01	Web	matchService.js	ISO 25010 Performance	Load test	Medium	Proposed
SWC-02	Reliability	Availability $\geq 99\%$ between 08:00–23:00.	Maintain service continuity.	System Architect	REQ-REL-01	Web	Firestore/Hosting	ISO 25010 Reliability	Uptime monitoring	High	Proposed

SWC-03	Security	No plaintext credentials; use Firebase Auth over HTTPS.	Protect user credentials.	Security Engineer	REQ-SEC-01	Web	authService.js	ISO 27001	Static analysis + pen test	High	Active
SWC-04	Privacy	Schedules visible only to user	Protect personal info.	Data Owner	REQ-PRIV-01	Web	Firestore rules	GDPR/KVKK	Access control tests	High	Proposed
SWC-05	Resource	Firestore schedule doc <1MB, ≤500 slots.	Stay within Firestore limits.	Backend Dev	REQ-DATA-01	Web	Firestore schema	Firestore Limits	Data model review	Low	Proposed

Notes

- These constraints are determined by software engineering requirements and belong in the Synkly Software Requirements Document (SRD).
- Software engineering validates behavior where it interacts with these constraints through testing, monitoring, and security audits.
- Each item listed here **must be traceable** to a Software Requirement in the Synkly SRD.

Traceability Reference

This document supports:

- (Synkly) Software Requirements Document (SRD)
- Product Assurance and Compliance Documentation
- MÜDEK Outcome: Interdisciplinary Integration & Constraint Awareness

Approval & Control

Role	Name	Signature	Date
Project Manager	Anıl Özişler		20.11.2025
Developer/Tester	Hatice Melike Aydın		20.11.2025
Developer/Requirements Analyst	İclal Zeynep Kaya		20.11.2025